HIROKI SAKAMOTO

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RESEARCH INTERESTS

- Model reduction of high-dimensional dynamical systems
- Model compression for deep learning models
- Machine learning and data science for modeling and control of complex systems

EDUCATION

Ph.D. Candidate in Mathematical Informatics, The University of Tokyo, Tokyo Apr. 2023 – Mar. 2026
Supervisor: Associate Professor Kazuhiro Sato

Exchange Student, Sorbonne University, Paris Sep. 2023 – Jan. 2024
Supervisor: Professor Emmanuel Trélat

M.S. in Mathematical Informatics, The University of Tokyo, Tokyo Apr. 2021 – Mar. 2023
Supervisor: Lecturer Kazuhiro Sato

B.S. in Mathematics, Waseda University, Tokyo Apr. 2017 – Mar. 2021
Supervisor: Professor Hitoshi Arai

RESEARCH POSITIONS

JSPS Research Fellow (DC2), Japan Society for the Promotion of Science Apr. 2025 – Present

PUBLICATIONS AND PREPRINTS

[1] H. Li, T. Nakata, **Hiroki Sakamoto**, and H. Uneya, “Lockdown policy rules with a hospital capacity constraint,” 2026, cARF Working Paper (CARF-F-620). [Online]. Available: https://www.carf.e.u-tokyo.ac.jp/wp/wp-content/uploads/2026/03/F620.pdf

[2] **Hiroki Sakamoto** and K. Sato, “Data-driven time-limited h^2 optimal model reduction for linear discrete-time systems,” 2026, under review at *IEEE Control Systems Letters*. [Online]. Available: https://arxiv.org/abs/2601.08372

[3] **Hiroki Sakamoto** and K. Sato, “A deep state-space model compression method using upper bound on output error,” 2025, under review at *IEEE Transactions on Automatic Control*. [Online]. Available: https://arxiv.org/abs/2510.14542

[4] **Hiroki Sakamoto** and K. Sato, “Compression method for deep diagonal state-space model based on $\mathcal{H}^2$ optimal reduction,” *IEEE Control Systems Letters*, vol. 9, pp. 2043–2048, 2025. [Online]. Available: https://ieeexplore.ieee.org/document/11087219

[5] **Hiroki Sakamoto** and K. Sato, “Data-driven h^2 model reduction for linear discrete-time systems,” 2024, preprint. [Online]. Available: https://arxiv.org/abs/2401.05774

[6] M. Obara, K. Sato, **Hiroki Sakamoto**, T. Okuno, and A. Takeda, “Stable linear system identification with prior knowledge by riemannian sequential quadratic optimization,” *IEEE Transactions on Automatic Control*, vol. 69, no. 3, pp. 2060–2066, 2024. [Online]. Available: https://ieeexplore.ieee.org/document/10258405

[7] **Hiroki Sakamoto** and K. Sato, “Stable probability of reduced matrix obtained by gaussian random projection,” *JSIAM Letters*, vol. 15, pp. 77–80, 2023. [Online]. Available: https://doi.org/10.14495/jsiaml.15.77

[8] **Hiroki Sakamoto** and K. Sato, “Random projection preserves stability with high probability,” *JSIAM Letters*, vol. 15, pp. 17–20, 2023. [Online]. Available: https://doi.org/10.14495/jsiaml.15.17

[9] **Hiroki Sakamoto** and K. Kawamoto, “Exploring factors driving young people’s migration from metropolitan to rural areas: An analysis using a modified gravity model,” *Tokei (Statistics)*, vol. 73, no. 3, pp. 56–63, 2022, (in Japanese).

CONFERENCE PRESENTATIONS

International Conferences

1. *Model compression for deep state space models using optimal-$\mathcal{H}^2$ model reduction*
   Hiroki Sakamoto, Kazuhiro Sato, University of Manchester and University of Tokyo Joint Research Symposium: Global Green Transformation Workshop, University of Manchester, UK, Oct. 2024

2. *$\mathcal{H}^2$ Model Order Reduction for Deep State-Space Models*
   Hiroki Sakamoto, Kazuhiro Sato, SICE Festival 2024 with Annual Conference, Kochi University of Technology, Japan, Aug. 2024

Conferences (Japan)

1. 出力誤差上界に基づく再学習なし Deep SSM の圧縮：LRA ベンチマークと WikiText-103 での検証
   坂本大樹, 佐藤一宏, The 13th SICE Multi-Symposium on Control Systems, 富山国際会議場, 2026 年 3 月

2. 出力誤差上界保証付き深層状態空間モデルの圧縮法
   坂本大樹, 佐藤一宏, 第 68 回自動制御連合講演会, 名古屋大学, 2025 年 11 月

3. $\mathcal{H}^2$ 最適モデル低次元化法による深層状態空間モデルの圧縮
   坂本大樹, 佐藤一宏, The 12th SICE Multi-Symposium on Control Systems, 大阪工業大学, 2025 年 3 月

4. 離散時間線形システムに対するデータ駆動型 h^2 モデル低次元化
   坂本大樹, 佐藤一宏, The 11th SICE Multi-Symposium on Control Systems, 広島大学, 2024 年 3 月

5. ランダム射影に基づく一般化シルベスター方程式 $AX + YB = C$ の高速求解法
   Hiroki Sakamoto, Pierre-Henri Tournier, Emmanuel Trélat, シンポジウム「錐線形計画と

その周辺」, 成蹊大学, 2024 年 2 月

6. ガウス埋め込みに基づく確率集中不等式の絶対定数の推定
   坂本大樹, 佐藤一宏, 日本応用数理学会第 19 回研究部会連合発表会, 岡山理科大学, 2023 年 3 月
7. ランダム射影による大規模なリヤプノフ方程式の高速求解法
   坂本大樹, 佐藤一宏, 日本応用数理学会 2022 年度年会, オンライン, 2022 年 9 月
8. 重力モデルによる若年層の大都市から地方への移動要因の解明
   川本晃大, 坂本大樹, 経済統計学会 2022 年度全国研究大会, オンライン, 2022 年 9 月

Other Talks

1. *Compression Method for Deep Diagonal State Space Model Based on $\mathcal{H}^2$ Optimal Reduction*
   坂本大樹, 佐藤一宏，制御理論若手合宿 2025, 広島, 2025 年 9 月
2. *Compression Method for Deep Diagonal State Space Model Based on $\mathcal{H}^2$ Optimal Reduction*
   Hiroki Sakamoto, Kazuhiro Sato, CSC Seminar, Max Planck Institute for Dynamics of Complex Technical Systems, Magdeburg, Germany, Aug. 2025
3. 制御理論の知見を活かした深層学習モデルのパラメータ削減
   坂本大樹, 佐藤一宏，情報学 for all by all, 東京大学, 2025 年 3 月
4. *Fast solving of generalized Sylvester matrix equations* $AX + YB = C$
   Hiroki Sakamoto, Pierre-Henri Tournier, Emmanuel Trélat, Workshop (Title TBD), University of Bologna, Italy, Dec. 2023
5. 感染と経済の統合モデル：Literature Review
   畝矢寛之, 川脇颯太, 坂本大樹, 芳賀沼和哉, 第 1 回コロナ政策研究会, 名古屋, 2022 年 9 月

GRANTS AND FELLOWSHIPS

JSPS Research Fellowship DC2, JSPS	Apr. 2025 – Mar. 2026
Overseas Study Support Program, JASSO	Sep. 2023 – Jan. 2024
Fellowship for Creation of Intelligent World, IIW, The University of Tokyo	Apr. 2023 – Mar. 2026
Outstanding Research Assistant, IIW, The University of Tokyo	Oct. 2022 – Mar. 2023
Grant for Independent Research, IIW, The University of Tokyo	Aug. 2022 – Mar. 2023

AWARDS

1. **システム制御情報学会学会賞奨励賞** 2026
2. **第 68 回自動制御連合講演会優秀発表賞** 2025
3. **統計データ分析コンペティション総務大臣賞** 2021

RESEARCH ASSISTANT AND INTERNSHIP

RA, Collaborative Project with Industry Aug. 2024 – Mar. 2025

Engaged in research and development of deep-learning models for industrial systems.

RA, Graduate School of Economics, The University of Tokyo Feb. 2022 – Feb. 2024

Development of a Healthcare Forecasting Tool for Japan using the SIR Model. For more details, see the project website.

Internship, Recruit Co., Ltd. Sep. 2022 – Dec. 2022

Improved the point-awarding algorithm for Hot Pepper.

TEACHING EXPERIENCE

TA, Dept. of Mathematical Engineering and Information Physics, The University of Tokyo Oct. 2022 – Feb. 2023

Supported a mathematical optimization class.

TA, Dept. of Mathematical Engineering and Information Physics, The University of Tokyo Apr. 2020 – Mar. 2021

Supported a C language programming exercise class.

Learning Assistant, Center for Data Science, Waseda University Apr. 2020 – Mar. 2021

Supported classes covering fundamentals of statistics, machine learning, and programming.

TECHNICAL SKILLS

- Programming languages: MATLAB, Python.
- Software: PyTorch, TensorFlow.
- Cloud Computing & GPU Acceleration: Built deep learning models on GPU-enabled VM instances on Google Cloud Platform.